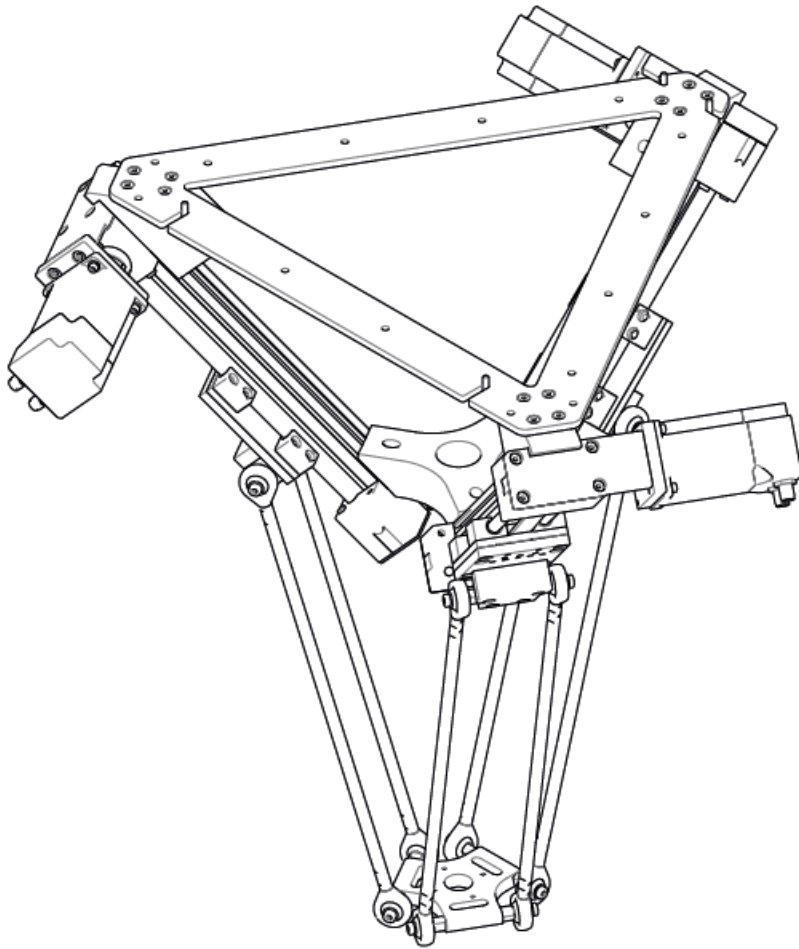




Parallel Manipulators: Kinematic Analyses of Delta 360 by IGUS

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OVERVIEW

This lecture note is prepared to show step by step kinematic analyses of an sTPM called Delta 360 by IGUS company. The actual robot is shown in Figure 1. The fixed platform that is used to fix the robot on a base structure and the moving platform are indicated in this figure. The moving platform can move linearly in all three Cartesian directions and hence, it is called an sTPM. The kinematic structure of this manipulator appears as $3PS^2S^2$ in the literature. However, it is mostly called a Delta manipulator with linear actuation.

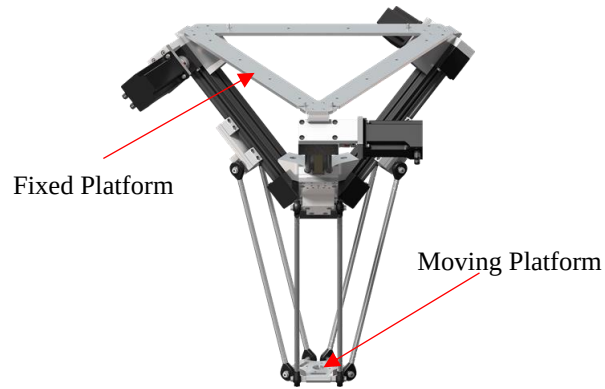


Figure 1. Fixed and moving platforms of Delta 360 manipulator

The major parts that are used to construct Delta 360 manipulator are shown in Figure 2 and they are available in the Technical Document of Delta 360 by IGUS.

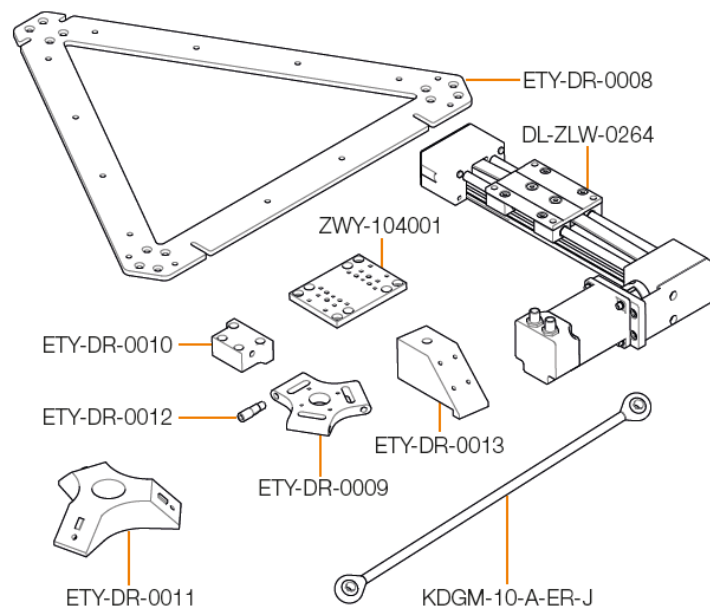


Figure 2. Mechanical components of Delta 360 manipulator by IGUS

ETY-DR-008: Fixed platform

ETY-DR-009: Moving platform

DL-ZLW-0264: Linear actuator

KDGM-10-A-ER-J: Link between the moving platform and fixed platform

ZWY-104001 and ETY-DR-0010 pieces are mounted on the slider of the linear actuator. On these parts two links are mounted via spherical joints. Then the other ends of these two links are mounted on the moving

platform via spherical joints. Hence, in one leg of the manipulator we have 1 prismatic and 4 spherical joint among which two of them are connected to the moving platform. This makes the configuration of a single leg PS^2S^2 .

DEFINITIONS OF THE DELTA 360 MANIPULATOR'S KINEMATICS

It should be mentioned that Delta 360 manipulator or in other words, PS^2S^2 manipulator, would have worked if we had a 3PUU configuration. 3PUU manipulator is also mentioned in the lecture notes on theory of parallel manipulators. The kinematic figure of 3PUU is shown in Figure 3.

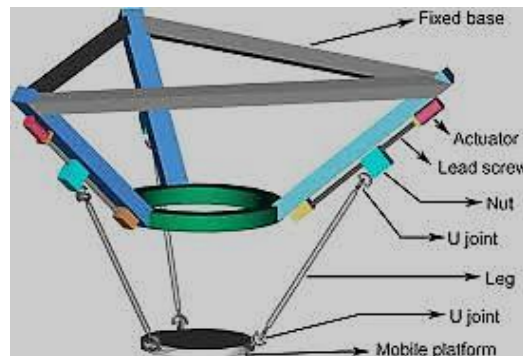


Figure 3. 3PUU spatial translational parallel manipulator

The reason to have double links in one leg of the Delta 360 is to increase the rigidity of the assembly. In terms of kinematics, Delta 360 manipulator is equivalent to 3PUU manipulator shown in Figure 3. Hence, we can assume that there is a single link right in between the two links of a leg. That means, instead of two links, we use a single virtual link for each leg. This virtual link is indicated with red lines in Figure 4. From the top view of the manipulator in Figure 4, it is seen that the angle between each leg is 120° . From the side view in Figure 4, it is seen that the angle between each linear actuator's axis of action and the fixed platform is 45° .

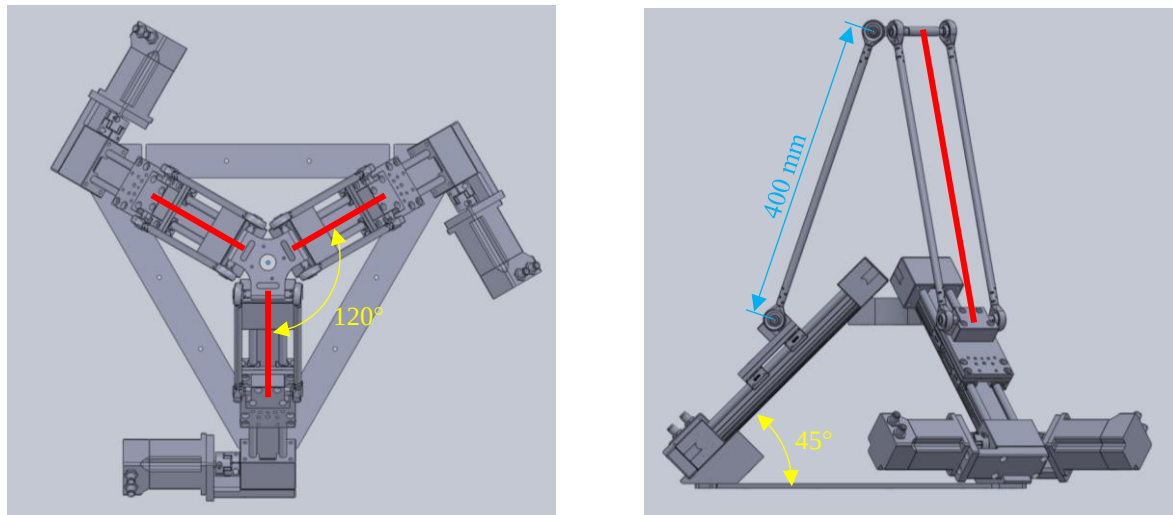


Figure 4. Virtual links inserted on the Delta 360 manipulator

The spherical joints (S) provide 3 relative rotational DoFs between two rigid bodies. Having two spherical joints at two ends of a link enables the link to rotate about the axis drawn from the center of one spherical joint to the other spherical joint's center. This type of motion is not necessary in terms of kinematics. Therefore, instead of two spherical joints at the two ends of the link, we can assume that there are two universal joints. Recall that universal joints (U) provide 2 relative rotational DoFs between two rigid bodies. Hence, having two U joints at

the two ends of the link constrains the rotation about the axis drawn from the center of one universal joint to the other universal joint's center.

In terms of kinematics, the link length of the virtual link is equal to the length between the centers of the spherical joints at both ends of link of the Delta 360 by IGUS, $l = 400$ mm. The link length to be used in kinematics is shown in Figure 4 on the side view picture printed in light blue color.

The dimensions of the fixed platform and the moving platform are given in the Technical Document of Delta 360 by IGUS. However, these dimensions are meaningless in terms of kinematics. In terms of kinematics, to identify the location of the virtual fixed platform (VFP), each actuator is retracted to the closest location to the actual fixed platform. Then, the VFP is placed by using the joint centers of the spherical joint that are connecting the links to the sliders of the linear actuators. The VFP is shown in Figure 5 with red colored lines.

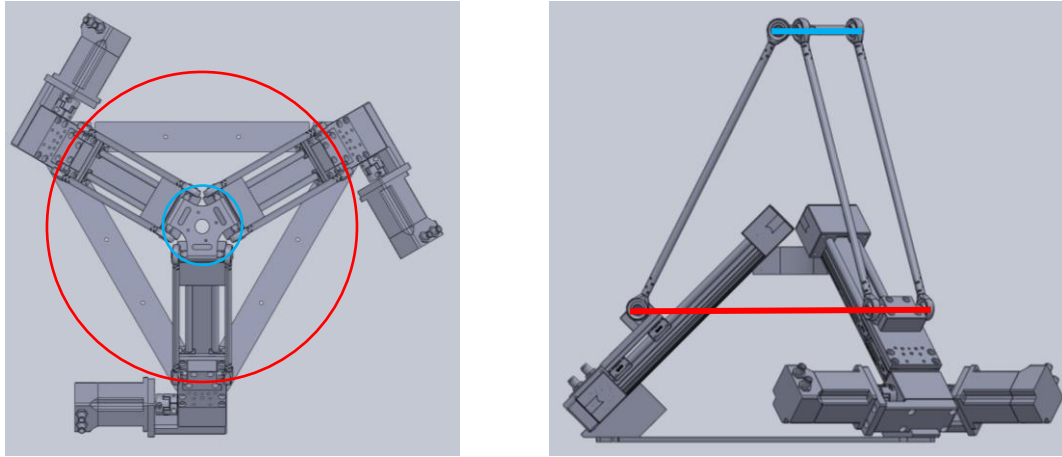


Figure 5. The location of the VFP for Delta 360 manipulator

Instead of the joint centers of the spheres, we could have chosen another location of the linear actuator or even a circle on the fixed platform to identify VFP. However, we would add extra constants in our kinematics in this way and it would not affect how the system works in terms of kinematics. Hence, the most convenient way to get rid of all the unnecessary constant parameters to appear in the kinematic analyses is to select the VFP as a circle passing from the centers of spherical joints that connect the links to the sliders of the actuators when the actuators are in their fully retracted positions. The fully retracted position of the actuators is when the sliders are 3 mm away from the casing, which is 1 mm away from the mechanical stoppers located on the motor casing.

The view in Figure 5 is obtained when the mechanical stoppers are removed from the linear actuators. Considering that the maximum travel of the sliders is up to 1 mm close to the mechanical stop on the other end as well, the linear actuators have a travel range of 144 mm. The radius of the VFP circle is then measured from the CAD file of the manipulator as $r_b = 222.105421$ mm.

A virtual moving platform (VMP) is defined in terms of its radius and location. Since the link length is defined based on the distance between the centers of the spherical joints located at both ends of the link, the VMP is defined by using the joint centers of the spherical joint that are connecting the links to the moving platform. The VMP is shown in Figure 5 with light blue colored lines. The radius of the VMP circle is then measured from the CAD file of the manipulator as $r_m = 42$ mm.

Based on the above-mentioned information on kinematics representation of Delta 360 manipulator, its kinematic sketch is drawn in Figure 6. On Figure 6, among the joint variables, only the primary variables (s_1, s_2, s_3) are denoted in red color. Due to the specific nature of the mechanism (sTPM), we will not need to solve for the secondary variables during the forward or inverse kinematic analyses. The primary variables are displacement of the linear actuators' sliders from the fully retracted positions defined in Figure 5.

In Figure 6, the construction parameters of the manipulator (r_b , r_m , l , φ , θ_b , θ_m) are identified with dark green color. Also, one of the three U-joints that are connected to the base platform and moving platform are identified in orange color.

The coordinate frames attached to the base platform (X - Y - Z) and attached to the moving platform (x - y - z) are parallel to each other throughout the manipulation. This is because Delta 360 manipulator is an sTPM and thus, the moving platform does not have any orientation change during the manipulation. The aim of driving the linear actuators is to re-position the center of the coordinate system (x - y - z) attached to the moving platform O_m with respect to the center of the coordinate system (X - Y - Z) attached to the base platform O_b .

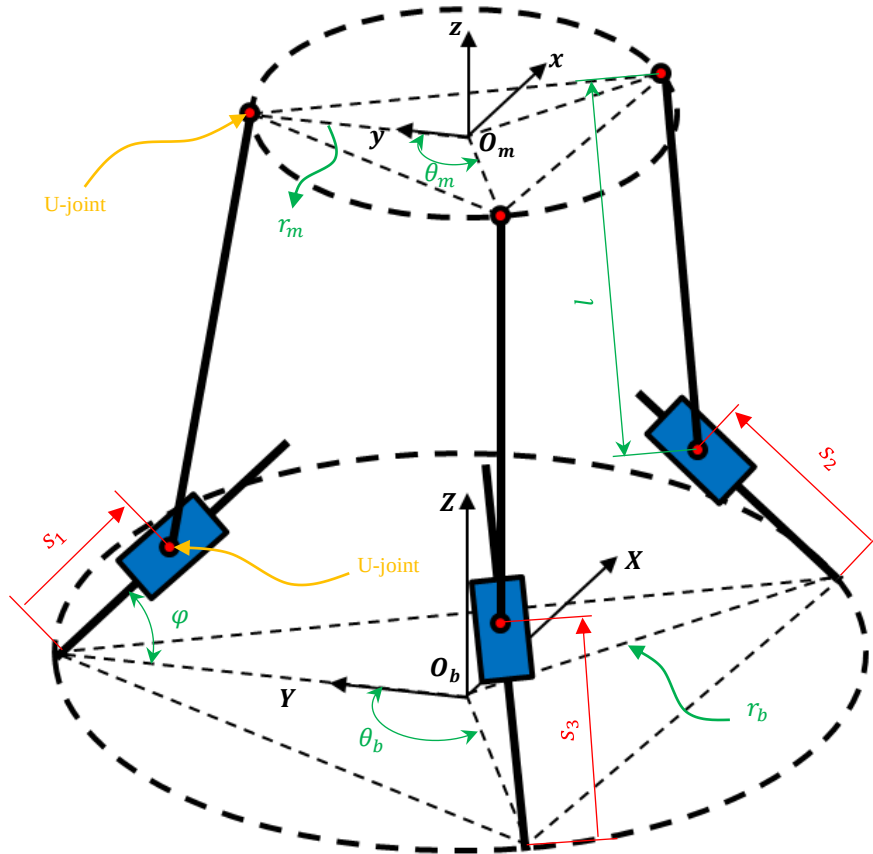


Figure 6. Reduced kinematic sketch of Delta 360 manipulator and its kinematic parameters

With the understanding of the kinematic parameters, we can now create a table informing us about the value of these parameters for Delta 360 manipulator by IGUS.

Table 1. Kinematic parameters of Delta 360 manipulator by IGUS

s_1, s_2, s_3	Primary variables: input variables supplied by the motion of the linear actuators	Range of motion of all linear actuators is 0 – 144 mm
r_b	Radius of the virtual fixed platform: this is defined by using the midpoint of the line connecting two adjacent spherical joints' centers when all the linear actuators are retracted back to $s_i = 0$ for $i = 1,2,3$	$r_b = 222.105421\text{mm}$
r_m	Radius of the virtual moving platform: this is defined by using the midpoint of the line connecting two adjacent spherical joints' centers located on the moving platform	$r_m = 42\text{ mm}$
l	Length of each link: this is defined as the distance between the centers of the spherical joints at both ends of the link	$l = 400\text{ mm}$

φ	The angle between the linear actuator's motion axis and the $\mathbf{X} - \mathbf{Y}$ plane of the VFP	$\varphi = 45^\circ$
θ_b	This is the relative angle between the two adjacent actuators as measured on the $\mathbf{X} - \mathbf{Y}$ plane of the VFP	$\theta_b = 120^\circ$
θ_m	This is the relative angle between the two adjacent U-joint as measured in $\mathbf{x} - \mathbf{y}$ plane of the VMP	$\theta_m = 120^\circ$